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FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

MOBILE DEVELOPMENT AND NETWORK ADMINISTRATION

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Which multiplex techniques are adapted in cellular systems and It covers what cell?	L2	Co1	**02**
	b.	Bluetooth technology communicates with each other at what frequency and maximum at what distance it can communicate?	L2	CO2	**02**
	c.	Which program coding written techniques is adopted in Android apps?	L3	CO3	**02**
	d.	Define the role of a database server in web application architecture.	L3	CO4	**02**
	e.	What is the difference between a file server and a database server?	L3	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Compare the Wireless Network with Wired Network?	L2	CO1	**05**
		OR			
	b.	List out the types of Computer Network Architecture	L2	CO1	**05**
3.	**a.**	Give details about Mobile Handset Characteristics (any 5)?	L3	CO2	**05**
		OR			
	b.	What are the functions and Usage of eSIM?.	L3	C02	**05**
4.	**a.**	Explain how would you would use an Integrated Development	L1	CO3	**05**
		OR			
	b.	How would you implement user authentication in a mobile app using Firebase in Android Studio?	L3	C03	**05**
5.	**a.**	How would you implement dynamic content on a website to improve user engagement? Provide a practical example.	L3	CO4	**05**
		OR			
	b.	Explain how an RSS feed can be integrated into a mobile application to provide up-to-date content. Demonstrate this with an example.	L1	C04	**05**
6.	**a.**	What are various Network Administrator's responsibilities	L2	CO5	**05**
		OR			
	b.	How does a network administrator handle network backups and disaster recovery planning?	L2	C05	**05**
7.	**a.**	Discuss the differences between dynamic, static, and automatic IP address allocation in DHCP?	L3	CO5	**05**
		OR			

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	b.	Explain the importance of network security in a network administrator's role and the measures they implement.	L3	C05	05
		PART-C		**6 X 10=60**	
8.	a.	Explain IP, MAC, PORT Address?	L2	CO1	10
		OR			
	b.	Explain the GSM Architecture with neat diagram?	L2	CO1	10
9.	a.	Describe the hardware architecture of a modern mobile handset. Discuss key components such as the CPU, memory, internal storage, and power management.	L3	CO2	10
		OR			
	b.	Discuss the role and importance of device drivers in mobile handsets. Explain how they interact with hardware components.	L3	C02	10
10.	a.	Describe the best practices for mobile app programming. Why are these practices important for the success of mobile applications?	L2	CO3	10
		OR			
	b.	Explain the mobile application development paradigm. How has mobile app development evolved from the early days to the current era?	L2	C03	10
11.	a.	How would you configure an Apache web server to serve a simple HTML webpage?	L3	CO4	10
		OR			
	b.	How would you evaluate the load-handling capabilities of a web server using load testing tools like Apache JMeter?	L3	C04	10
12.	a.	Discuss the importance of DNS security and explain how DNSSEC works to protect DNS queries from attacks. What are the limitations of DNSSEC?	L3	CO5	10
		OR			
	b.	Explain the structure of the Domain Name System (DNS). How does DNS translate domain names into IP addresses?	L3	C05	10
13.	a.	Use the nslookup or dig command to perform DNS queries. How would you apply these commands to troubleshoot DNS issues and retrieve domain information?	L3	CO5	10
		OR			
	b.	Demonstrate the use of the ifconfig or ip command to configure and manage network interfaces on a Linux system. How would you apply these commands to check IP addresses, enable/disable interfaces, and modify IP settings?	L3	C05	10